

Capstone Group 19

Progress Report 2: Design Inputs

A Monitoring System for Tracking Parkinson's Tremor Over Time

IBEHS 5P06 Capstone

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1 Updated User and Problem Framing

1.1 Changes Since Report 1

Following PR1 feedback, monitoring Parkinson's rest tremor is the core deliverable, with posture and activity analysed separately. We are continuing to investigate electrical stimulation as an option. Parkinson Canada remains the stakeholder. FDA evidence guides our measures and missing-data reporting. It does not establish our monitor's performance.

1.2 User Priorities and Intended Benefit

The proposed daily timeline will summarize tremor fraction, detected frequency, recording coverage and user-entered context, supported by guided rest/posture measurements. Analysis will use non-overlapping three-second windows. Valid windows require complete, correctly timed, unclipped on-wrist data; these quality checks remain unverified. Detected tremor means a valid window passing the provisional signal gate, not a clinical diagnosis. Within each labelled context, tremor fraction is detected-window duration divided by valid-window duration. Valid gate-negative windows count as no detected tremor; invalid windows are excluded and flagged. Coverage is valid time divided by scheduled recording time. Frequency is displayed only for detected windows. Usefulness to patients and clinicians and acceptable recording effort still need evidence.

Need statement: Patients and clinicians need an understandable record of tremor patterns over time, supported by comparable guided measurements, to discuss symptom changes.

2 Design Inputs

2.1 Requirements and Traceability

Table 1 links needs to measurable inputs and verification. The initial software test target is provisional. Other acceptance limits require evidence on measurement accuracy and user needs before formal verification. Need IDs: N4 safe use; N5 comfort; N6 useful trends; N7 affordability; N8 control of personal data.

Table 1. Design inputs and their evidence

ID	Input and measurable criterion	Need or source	Verification
DI-01 Functional	Each summary includes time, estimated tremor fraction, frequency, context and coverage. Unavailable values are labelled; frequency is shown only when validity and detection checks pass.	N6; R-01; team proposal	Replay labelled intervals and check displayed values, missing-data states and detection status.
DI-02 Performance	Proposed frequency error in software tests ≤0.25 Hz (half the 0.5 Hz search spacing), for clean 4-6 Hz sinusoids only. Detection limits require motion data with known tremor status.	N6; software resolution target, not clinical accuracy	Independent 4-6 Hz signed-sine sweep, 20°/s peak, varied phases. Separately test noise and labelled ordinary/no-tremor motion.
DI-03 Performance	100% injected gaps flagged; valid-time denominator excludes every gap.	N6; R-01; informed by [2]	Known-gap replay; compare computed coverage.

ID	Input and measurable criterion	Need or source	Verification
DI-04 Interface	100% guided comparisons retain task/wrist labels; no silent pooling.	N6; R-01; team proposal	Audit labelled rest/posture comparisons.
DI-05 Usability	Setup time (min), assistance prompts and, for worn options, mass (g), wear time (h) and discomfort. Acceptance limits need user evidence.	N4/N5; R-02; usability evidence still needed	Check setup time and help prompts in software walkthroughs. Use published findings or measurements for worn options.

2.2 Constraints and Trade-Offs

Budget, hardware and wear-time limits remain unagreed. Compare concepts using estimated parts cost (CAD), recording time between charges (h), setup time (min), daily user actions and replaceability of the strap and battery. Evaluate sampling versus battery use and context detail versus interaction burden. Assess one-handed operation, caregiver assistance and electronic waste during concept comparison. Privacy input (N8): sharing must require an explicit user action; verify with a no-consent export test.

3 Concept Development Process and Preliminary Analysis

3.1 Process and Supervisor Consultation

Generate monitoring concepts using existing wearables (e.g., Apple Watch, Samsung Watch, WHOOP), phone measurements or custom sensing. Check documentation for access to suitable motion data. Compare feasible options by data quality, comfort and practicality, setup, cost and development effort. Evaluate the leading approach using evidence from literature, simulation or suitable recordings, as appropriate. Record uncertainty and revisit the choice as evidence develops. The current requirements assume wrist data and would need revision for another placement.

3.2 Feasibility Evidence and Uncertainty

Cala ONE missed its primary spiral endpoint ($p = .263$), but received De Novo authorization following benefit-risk review [3]. Cala kIQ's PD indication concerns postural/kinetic hand tremor [1]. Felix's ET review emphasized missing-data analysis [2, p. 21]. These findings motivate task-specific evaluation, prespecified outcomes and recording coverage; they do not establish our monitor's performance.

Inherited PADS tables and code are available, but raw-data recovery and reproduction remain pending. Its short clinic tasks [4] can support guided-task evaluation but cannot alone validate daily trends.

An October 23 host test of the unchanged features.c code used 624 samples at 208 Hz per window. Signed, noise-free sinusoids at 4, 5, 6, 4.25 and 5.75 Hz, each at three phases and 20 degrees/s peak amplitude, produced 15 cases. Frequency error was zero for the nine grid-aligned cases and 0.25 Hz for the six between-bin cases; the gate accepted all 15. It rejected zero-input and 1.2 Hz sine controls, although both returned a nominal 3 Hz estimate. These function-level results support detection-gated display; they do not validate sensor preprocessing, ordinary-motion rejection or clinical accuracy. Next, evaluate noise and ordinary movement within the proposed software proof of concept.

The initial proof of concept will process motion data and display tremor measurements, their context and whether the data are usable. Evaluation may use simulated signals or suitable recordings, with data quality, units and sampling checked before analysis. Methods can be refined as the concept develops. We will explain what each test shows and what remains untested.

4 Preliminary Risk Analysis

4.1 Hazards and Design Implications

Table 2 gives preliminary likelihood and severity assessments. The design, use conditions and effectiveness of the controls remain uncertain. Risks that could cause serious harm receive priority. Any stimulation work requires a separate risk review before wearer testing.

Table 2. Preliminary risks and their implications for design

Risk and context	Potential consequence	Control or evidence needed	Effect on concepts
R-01 Misleading timeline	Likelihood of harm is uncertain; possible causes include gaps, false detections and task changes. Severity could be serious if users act on misleading results.	Show coverage and context; suppress unsupported frequency values; verify missing-data and no-tremor cases.	Prioritize interpretability; avoid diagnostic claims.
R-02 Wear and power	For worn concepts, likelihood depends on fit, contact materials and power protection and remains uncertain. Potential severity: reversible skin irritation or serious injury from overheating.	If we later build a device, verify power protection, enclosure integrity, contact materials and wear limits before wearer testing.	For worn options, prioritize overheating protection alongside R-01, followed by fit and skin checks.

4.2 Standards and Ethical Considerations

ISO 14971:2019 is a proposed framework for hazard identification, controls and residual-risk review [5]; compliance has not been established. Confirm research ethics and consent requirements under TCPS 2 [6]. Assess the applicability of Canada's Medical Devices Regulations and any investigational-testing authorization requirements using the intended use and device classification [7]. U.S. stimulator clearances do not establish Canadian authorization for this monitor. Any future participant testing would require a separate plan and applicable approvals. Restrict access to identifiable data, define retention and deletion, and exclude diagnostic, staging and medication-advice claims.

References

[1] U.S. Food and Drug Administration, ‘Cala kIQ - 510(k) summary’, U.S. Food and Drug Administration, Nov. 2022. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf22/K222237.pdf

[2] U.S. Food and Drug Administration, ‘Felix NeuroAI - 510(k) summary’, U.S. Food and Drug Administration, Jul. 2025. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/pdf25/K250096.pdf

[3] U.S. Food and Drug Administration, ‘Cala ONE - De Novo classification review’, U.S. Food and Drug Administration, Apr. 2018. [Online]. Available: https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN170028.pdf

[4] J. Varghese, A. Brenner, L. Plagwitz, C. van Alen, M. Fujarski, and T. Warnecke, ‘PADS - Parkinson’s Disease Smartwatch dataset (version 1.0.0)’. PhysioNet, Mar. 25, 2024. doi: 10.13026/m0w9-zx22.

[5] International Organization for Standardization, ‘ISO 14971:2019 - Medical devices: Application of risk management to medical devices’, public standard description, 2019. Accessed: Oct. 23, 2026. [Online]. Available: https://www.iso.org/standard/72704.html

[6] Canadian Institutes of Health Research, Natural Sciences and Engineering Research Council of Canada, and Social Sciences and Humanities Research Council of Canada, ‘Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans - TCPS 2 (2022)’, 2022. [Online]. Available: https://ethics.gc.ca/eng/documents/tcps2-2022-en.pdf

[7] Government of Canada, ‘Medical Devices Regulations’, SOR/98-282, Part 3, s. 80. Accessed: Oct. 23, 2026. [Online]. Available: https://laws-lois.justice.gc.ca/eng/regulations/SOR-98-282/section-80.html